

THE FRUIT PREDICTOR

Your First Machine Learning Model

Building Intelligent Brains with **scikit-learn**

Planrs Academy | Grade 6 Session 1

THE MYSTERY FRUIT

Can You Guess What This Is?

Imagine you are a detective. A suspect hands you a fruit, but you're blindfolded! You have to use your other senses to identify it. Here are your clues:

- **Rough, bumpy skin**
- **Many black seeds inside**
- **Creamy, sweet taste**

The Answer: It's a **Sitaphal** (Custard Apple)! How did your brain know that? Because you've seen, smelled, and tasted Sitaphals before. You learned from examples!

Machine Learning is exactly like this. Instead of teaching a computer "rules," we show it "examples."

DEFINING MACHINE LEARNING

Learning from Examples (Data)

In traditional coding, we give the computer a set of rules (If X, then Y). In **Machine Learning (ML)**, we give the computer data and let it find the patterns itself.

The Mango Variety Challenge:

Imagine teaching someone the difference between an **Alphonso** and a **Langra** mango. You don't just give a definition. You show them 50 of each variety. They look at the shape, smell the aroma, and check the color.

Soon, their brain creates a "pattern." When they see a 51st mango, they can predict which variety it is. That's ML!

How Computers "See" Objects

To a computer, a fruit is just a collection of **Features**. Features are characteristics that help us tell things apart.

Feature 1: Weight

How heavy is it? Heavier fruits might be juicier.

Feature 2: Texture

Is the skin smooth like a banana or rough like a pineapple?

Feature 3: Color

Green, Yellow, or Red? Color gives vital clues about ripeness.

The Sabzi Mandi Expert

Think about when you go to the **Vegetable Market (Sabzi Mandi)** with your parents to buy a watermelon. How do they pick the best one?

The Watermelon Test:

- They check the **Weight** (Feature 1).
- They listen to the **Tapping Sound** (Feature 2).

Your parents are acting like a Machine Learning model! They use these clues to **Predict** if the fruit is sweet and red inside.

THE MACHINE LEARNING CYCLE

Train, Test, Predict

When we build a model with a tool like **scikit-learn**, we always follow these three steps:



TRAIN

We show the computer labeled examples (e.g., "This is a Mango").



TEST

We show it a new fruit and check if the computer gets it right.



PREDICT

We let the "trained" computer guess fruits it has never seen before!

The Brain Builder

scikit-learn is a powerful library for Python that acts like a "Brain Builder." It provides the tools to take your data (the fruit examples) and build a **Decision Tree**.

What is a Decision Tree?

Imagine a series of "Yes/No" questions:

1. Is the fruit heavy? (Yes)
2. Is the skin yellow? (Yes)
3. **Prediction:** It's a Banana!

scikit-learn automatically creates these trees based on the data you provide.

ORGANIZING OUR CLUES

The Fruit Data Table

Before the computer can learn, we have to organize our features into a table. This is our **Vishleshan** (Analysis) phase.

Weight (grams)	Texture (1=Smooth)	Label (The Fruit)
150	1	Banana
300	0	Mango

The computer looks at this table and learns that "150g + Smooth = Banana."

THE ANUMAAN PHASE

Predicting the Unknown

Once our model is trained, it's time for an **Anumaan** (Prediction). We give the model new features, and it gives us a guess.

The Test Case:

New Fruit: 310 grams, Bumpy Texture.

Model says: "Based on my training, I am 95% sure this is a Mango!"

The more examples (rows) we have in our table, the smarter and more accurate our model becomes.

SUMMARY

Your ML Journey Starts Here

- **Machine Learning:** Computers learning from patterns, not just rules.
- **Features:** The "Clues" like weight or color.
- **scikit-learn:** The tool that builds the digital "Brain."
- **The Cycle:** Train, Test, and then Predict.

Success Criteria

Can you explain the 3 steps of ML? Can you identify 2 features of your favorite snack? If yes, you are a Young ML Explorer!